

Topic 2: Multiple Regression Analysis
Chapter 13 of our Textbook

STAT 2509: Introduction to Statistical Modeling II

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Multiple Linear Regression (MLR): Overview

Extension of Simple Linear Regression:

- One response variable y
- Multiple predictor variables $x_1, x_2, \dots, x_k$
- Applications: Economics, biology, engineering, social sciences
- Goal: Model relationship and make predictions

Example: Modeling house prices based on:

- x_1 : Square footage
- x_2 : Number of bedrooms
- x_3 : Age of house
- x_4 : Location rating

Model Formula and General Linear Model

Multiple Linear Regression (MLR) Model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i$$

- y_i : Response variable for observation i
- x_{ij} : Value of j -th predictor ($j = 1, 2, \dots, k$) for observation i ($i = 1, 2, \dots, n$)
- β_0 : Intercept (expected y when all $x_j = 0$)
- β_j : Partial regression coefficient (slope) for x_j
- ε_i : Random error term

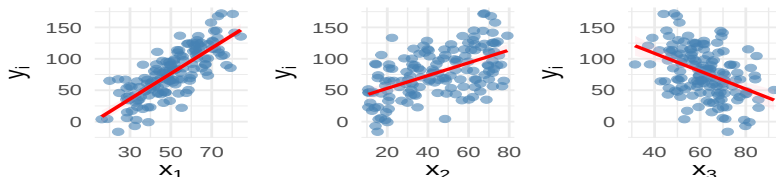
Matrix Form: (Not used in this course) $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

Assumptions of Multiple Linear Regression

- 1 **Linearity:** Relationship between y and x 's is linear

$$\text{Linear Relationships: } y_i = \beta_0 + \sum_{j=1}^3 \beta_j x_{ij} + \varepsilon_i$$



If each explanatory variable exhibits an approximate linear relationship, then a model of the form

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

would be useful for explaining and predicting y .

- 2 **Independence:** Errors ε_i are independent
- 3 **Homoscedasticity:** Constant variance $\text{Var}(\varepsilon_i) = \sigma^2$
- 4 **Normality:** $\varepsilon_i \sim N(0, \sigma^2)$
- 5 **No perfect multicollinearity:** Predictors not perfectly correlated
- 6 **Fixed predictors:** x 's are fixed and measured without error

Checking assumptions: Residual plots, diagnostic tests

Assumptions of Multiple Linear Regression

As in SLR model, observations of $x_1, \dots, x_k$ are treated as non-random constant.
Hence by MLR model

$$y = \underbrace{\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k}_{\text{constant}} + \underbrace{\varepsilon}_{\sim N(0, \sigma^2)}$$

as x 's are measured without errors and $\varepsilon \sim N(0, \sigma^2)$. We have

$$y \sim N\left(\underbrace{\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k}_{E(y)}, \underbrace{\sigma^2}_{\text{var}(y)}\right)$$

i.e., y has a normal distribution with mean

$$E(y) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k \quad \text{and variance} \quad V(y) = \sigma^2.$$

In general, a model qualifies as a multiple linear regression model if, apart from an additive error term ε , each additive component is linear in its β -parameter(s).

The following examples can be seen as MLR models

- Interaction term:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 \underbrace{x_1 x_2}_{x_3} + \varepsilon$$

- Polynomial terms:

$$y = \beta_0 + \beta_1 \underbrace{x}_{x_1} + \beta_2 \underbrace{x^2}_{x_2} + \beta_3 \underbrace{x^3}_{x_3} + \varepsilon$$

- MLR with higher order and interaction terms:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 \underbrace{x_1^2}_{x_3} + \beta_4 \underbrace{x_1 x_2}_{x_4} + \varepsilon$$

- Trigonometric transformation:

$$y = \beta_0 + \beta_1 \underbrace{x}_{x_1} + \beta_2 \underbrace{\sin(x)}_{x_2} + \varepsilon$$

- MLR with exponential variable transformation:

$$y = \beta_0 + \beta_1 \underbrace{x}_{x_1} + \beta_2 \underbrace{e^x}_{x_2} + \varepsilon$$

Note: A term like x^2 or x^3 , etc is called a **higher order term**. A term like $x_1 x_2$ is called an **interaction term**.

The following examples are not seen as MLR models (Non-linear in regression parameters)

- Exponential model:

$$y = e^{\beta_0 + \beta_1 x} + \varepsilon$$

- Rational model with multiplicative error:

$$y = \frac{1}{(\beta_0 + \beta_1 x)} \varepsilon$$

Method of Least Squares (LS)

- Similar to the SLR case, the least-squares (LS) estimators of $\beta_0, \beta_1, \dots, \beta_k$ are found by minimizing the sum of squared fitting errors.
- We denote the LS estimator of β_j as $\hat{\beta}_j, j = 0, 1, \dots, k$.
- It can be shown that $\hat{\beta}_j$ is an unbiased estimator of β_j , i.e. $E(\hat{\beta}_j) = \beta_j$, for all $j = 0, 1, \dots, k$.
- The formula for calculating the LS estimators $\hat{\beta}_j$ is not required. We directly obtain $\hat{\beta}_j$ using R.
- After obtaining $\hat{\beta}_j, j = 0, 1, \dots, k$, the least-squares regression line (or, prediction line, fitted regression line) is given by

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_k x_k.$$

- $\hat{y}$ is called the fitted value (or, predicted value) of y .
- **Residuals:** $e_i = y_i - \hat{y}_i, i = 1, 2, \dots, n$.

Example Dataset ($n = 15$)

Research Question: Predict final exam score based on:

- x_1 : Hours studied per week
- x_2 : Average assignment score
- y : Final exam score (out of 100)

Student	Hours/week (x_1)	Assign. score (x_2)	Final (y)
1	10	75	68
2	12	80	75
3	8	70	62
$\vdots$	$\vdots$	$\vdots$	$\vdots$
15	14	85	82

R lm() Output and Interpretation

```
> model <- lm(Final ~ Hours + Assignment, data=exam)
> summary(model)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	15.3421	8.7214	1.759	0.1034
Hours	2.1563	0.4823	4.471	0.0007 ***
Assignment	0.4872	0.1031	4.725	0.0005 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.892 on 12 degrees of freedom

Multiple R-squared: 0.8724, Adjusted R-squared: 0.8512

F-statistic: 41.07 on 2 and 12 DF, p-value: 1.234e-06

Regression equation:

$$\widehat{Final} = 15.34 + 2.16(Hours) + 0.49(Assignment)$$

Interpretation of $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$

- **Intercept $\hat{\beta}_0$:** The expected value of y when all predictors $x_1, x_2, \dots, x_k$ are equal to zero.
- **Slope $\hat{\beta}_j$ ($j = 1, \dots, k$):** The expected change in y for a one-unit increase in x_j , *holding all other predictors constant*.

Example: Suppose

$$\hat{y} = 50 + 3x_1 - 2x_2,$$

where x_1 is years of education and x_2 is hours of absence.

- $\hat{\beta}_1 = 3 > 0$: Each additional year of education increases the average y by 3 units, holding x_2 constant.
- $\hat{\beta}_2 = -2 < 0$: Each additional hour of absence decreases the average y by 2 units, holding x_1 constant.

Example from R: Interpretation of $\hat{\beta}_1$ and $\hat{\beta}_2$

- $\hat{\beta}_1 = 2.16$: Each additional hour increases the average final score by 2.16 points, holding assignment score constant
- $\hat{\beta}_2 = 0.49$: Each additional assignment point increases the average final by 0.49 points, holding study hours constant

Partitioning of Variation (Similar to SLR)

- Total sum of squares (*Total variation*) is given by

$$TSS = S_{yy} = \sum (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2 = \sum y_i^2 - \frac{(\sum y_i)^2}{n}.$$

- Sum of squared errors (*Unexplained variation*) is given by

$$SSE = \sum (y_i - \hat{y}_i)^2.$$

- Sum of squares for regression (*Explained variation*) is given by

$$SSR = \sum (\hat{y}_i - \bar{y})^2.$$

- Total variation is the sum of explained and unexplained variation:

$$TSS = SSR + SSE$$

Degrees of Freedom:

- $df(SSR) = k$
- $df(SSE) = n - k - 1$
- $df(TSS) = n - 1$

Analysis of Variance (ANOVA) Table for Multiple Linear Regression

Table: The analysis of variance (ANOVA)

Source	SS	df	MS	F	Rejection region
Regression	SSR	k	$MSR = \frac{SSR}{k}$	$\frac{MSR}{MSE} = \frac{SSR/k}{SSE/(n-k-1)}$	
Error	SSE	$n - k - 1$	$MSE = \frac{SSE}{n - k - 1}$		
Total	TSS	$n - 1$			

Note:

- If $k = 1$, the above values in the ANOVA table are the same as those in SLR.
- The estimator of σ^2 is $s^2 = MSE = \frac{SSE}{n-k-1}$.

F-test for overall model significance

Test $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ (regression is bad) vs $H_a : \beta_i \neq 0$ for some i (regression is useful)

$$F = \frac{MSR}{MSE} = \frac{SSR/k}{SSE/(n-k-1)} \sim F(k, n-k-1) \quad \text{under } H_0$$

Reject H_0 if $F > F_\alpha(k, n-k-1)$ or p-value $< \alpha$, where α is the significant level.

Example from R: Testing the Significance of the Overall Model

$H_0 : \beta_1 = \beta_2 = 0$ (regression is bad) vs $H_a : \beta_i \neq 0$ for some i (regression is useful)

Coefficients:

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Residual standard error: 3.892 on 12 degrees of freedom

Multiple R-squared: 0.8724, Adjusted R-squared: 0.8512

F-statistic: 41.07 on 2 and 12 DF, p-value: 1.234e-06

We compare the test statistic $F = 41.07$ with the critical value from the F -distribution at the significance level $\alpha = 0.05$, with degrees of freedom $df_1 = 2$ and $df_2 = 12$. The critical value is $F_{0.05,2,12} = 3.8853$. Since $41.07 > 3.8853$, we reject the null hypothesis H_0 and conclude that the model is useful for predicting the final score.

Alternatively, we compare the p -value with the significance level $\alpha = 0.05$. Because the p -value is $1.234 \times 10^{-6} < 0.05$, we again reject H_0 and conclude that the model is useful for predicting the final score.

Testing Partial Regression Coefficients

For each coefficient $\hat{\beta}_j \sim N(\beta_j, \text{Var}(\hat{\beta}_j))$ where $j = 1, 2, \dots, k$:

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_a : \beta_j \neq 0$$

Test Statistic:

$$t = \frac{\hat{\beta}_j - 0}{\text{s.e.}(\hat{\beta}_j)} \sim t_{n-k-1} \text{ under } H_0$$

where $\text{s.e.}(\hat{\beta}_j) = \sqrt{\text{Var}(\hat{\beta}_j)}$ is the standard error of $\hat{\beta}_j$ obtained from R.

We reject H_0 in favor of H_a at the level of significant α if $t \geq t_{\alpha/2; n-k-1}$ or $t \leq -t_{\alpha/2; n-k-1}$, or if $p\text{-value} \leq \alpha$.

Confidence Interval:

$$\hat{\beta}_j \pm t_{\alpha/2, n-k-1} \cdot \text{s.e.}(\hat{\beta}_j)$$

Example from R:

- For β_1 (Hours): $t = \frac{2.1563}{0.4823} = 4.471$, $p = 0.0007$
- For β_2 (Assignment): $t = \frac{0.4872}{0.1031} = 4.725$, $p = 0.0005$
- Both significant at $\alpha = 0.05$

95% CI for β_1 :

$$2.1563 \pm (2.179 \times 0.4823) \approx [1.105, 3.208]$$

95% CI for β_2 :

$$0.4872 \pm (2.179 \times 0.1031) \approx [0.263, 0.712]$$

Comparing Nested Models (Partial F-test)

Suppose the **full model** with k explanatory variables:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \varepsilon$$

Let $\beta_{l_1}, \beta_{l_2}, \dots, \beta_{l_{k-g}}$ be a subset of $\beta_1, \dots, \beta_k$.

For example, if $k = 10$ and $g = 7$ (so $k - g = 3$), then $\beta_{l_1}, \dots, \beta_{l_{k-g}}$ can be $\beta_3, \beta_4, \beta_5$ or $\beta_1, \beta_6, \beta_9$.

Suppose that we want to test the hypotheses:

$$H_0 : \beta_{l_1} = \beta_{l_2} = \cdots = \beta_{l_{k-g}} = 0$$

vs H_a : at least one of $\beta_{l_1}, \beta_{l_2}, \dots, \beta_{l_{k-g}}$ is not 0

To test the hypotheses, we do the following steps:

- 1 Form a **reduced model** by removing all terms corresponding to $\beta_{l_1}, \beta_{l_2}, \dots, \beta_{l_{k-g}}$ from the full model.

For example, if the **full model** is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_6 x_6 + \varepsilon$$

and the H_0 is $H_0 : \beta_1 = \beta_2 = \beta_5 = 0$, then the **reduced model** is

$$y = \beta_0 + \beta_3 x_3 + \beta_4 x_4 + \beta_6 x_6 + \varepsilon,$$

which is a submodel of the full model.

- 2 Fit the full model and the reduced model separately to the same data.
- 3 Denote the SSE and SSR for the full model as SSE_f and SSR_f , respectively, and the SSE and SSR for the reduced model as SSE_r and SSR_r , respectively.

For testing

$$H_0 : \beta_{l_1} = \beta_{l_2} = \cdots = \beta_{l_{k-g}} = 0 \quad (\text{Reduced model})$$

$$\text{vs } H_a : \text{at least one of } \beta_{l_1}, \dots, \beta_{l_{k-g}} \text{ is not } 0 \quad (\text{Full model}),$$

we use the so called **partial- F -test** statistic defined as

$$F_{\text{drop}} = \frac{(SSE_r - SSE_f)/(df_{SSE_r} - df_{SSE_f})}{SSE_f/df_{SSE_f}}$$

$$= \frac{(SSE_r - SSE_f)/(k - g)}{SSE_f/(n - k - 1)} \sim F_{(k-g), (n-k-1)}$$

where df_{SSE_r} and df_{SSE_f} are the degrees of freedom of SSE_r for the reduced model and the full model respectively.

Note that $df_{SSE_r} = n - g - 1$ and $df_{SSE_f} = n - k - 1$, so

$$df_{SSE_r} - d.f._{SSE_f} = (n - g - 1) - (n - k - 1) = k - g.$$

We reject H_0 in favor of H_a at the α th significant level if $F_{\text{drop}} \geq F_{\alpha; (k-g), (n-k-1)}$.

Full Model: $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$ vs **Reduced Model:** $y = \beta_0 + \beta_1 X_1 + \varepsilon$
Hypotheses: $H_0 : \beta_2 = 0$ (Reduced model) vs $H_a : \beta_2 \neq 0$ (Full model)

Example from R:

```
> reduced <- lm(Final ~ Hours, data=exam)
> full <- lm(Final ~ Hours + Assignment, data=exam)
> anova(reduced, full)
```

```
Model 1: Final ~ Hours
Model 2: Final ~ Hours + Assignment
Res.Df    RSS Df Sum of Sq      F      Pr(>F)
1      13 519.94
2      12 181.73   1    338.21 22.329 0.0004732 ***
```

From the R output:

- Reduced model ($\text{Final} \sim \text{Hours}$): $\text{SSE}_r = 519.94$, $df_{\text{SSE}_r} = 13$
- Full model ($\text{Final} \sim \text{Hours} + \text{Assignment}$): $\text{SSE}_f = 181.73$, $df_{\text{SSE}_f} = 12$
- Extra sum of squares: $\text{Extra SS} = 519.94 - 181.73 = 338.21$, $\text{Extra df} = 1$

The partial F -statistic is:

$$F_{\text{drop}} = \frac{(\text{SSE}_r - \text{SSE}_f) / (df_{\text{SSE}_r} - df_{\text{SSE}_f})}{\text{SSE}_f / df_{\text{SSE}_f}} = \frac{(519.94 - 181.73) / (13 - 12)}{181.73 / 12} = \frac{338.21}{15.144167} \approx 22.329$$

This matches the R output: $F = 22.329$, $p\text{-value} = 0.0004732$.

Conclusion: Reject H_0 . There is strong evidence that the **Assignment** coefficient is not zero, after accounting for **Hours**.

A note on the F_{drop} test statistic of the partial F-test:

Under both the full and the reduced models

$$TSS_f = \sum_{i=1}^n (y_i - \bar{y})^2 = TSS_r$$

Note TSS is determined by y_i 's only. Using $SSE = TSS - SSR$, we have

$$SSE_r = TSS_r - SSR_r, \quad \text{and} \quad SSE_f = TSS_f - SSR_f.$$

$$\Rightarrow \boxed{SSE_r - SSE_f = SSR_f - SSR_r}.$$

In addition

$$df_{SSR_f} - df_{SSR_r} = k - g = df_{SSE_r} - df_{SSE_f}.$$

Hence

$$\begin{aligned} F_{\text{drop}} &= \frac{(SSE_r - SSE_f)/(df_{SSE_r} - df_{SSE_f})}{SSE_f/df_{SSE_f}} \\ &= \frac{(SSE_r - SSE_f)/(k - g)}{SSE_f/(n - k - 1)} \\ &= \frac{(SSR_f - SSR_r)/(df_{SSR_f} - df_{SSR_r})}{SSE_f/(n - k - 1)} \\ &= \frac{(SSR_f - SSR_r)/(k - g)}{SSE_f/(n - k - 1)} = F_{\text{part}} \end{aligned}$$

Where k is the number of explanatory variables in the full model and g is the number of explanatory variables in the reduced model. You can use any one of the above four equivalent formulas for the partial F -test.

Remark: Partial F test can be used to test hypotheses about a single β_j ($j = 1, \dots, k$)

Example: Full model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

Testing problem:

$$H_0 : \beta_2 = 0 \quad \text{vs} \quad H_a : \beta_2 \neq 0$$

For this problem, the reduced model is

$$y = \beta_0 + \beta_1 x_1 + \beta_3 x_3 + \varepsilon.$$

One then can use the partial F -test procedure to carry out the hypothesis test.

For testing

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_a : \beta_j \neq 0, \quad \text{for a given } j = 1, \dots, k,$$

we can use either t -test or partial F test.

Denote the student t -test and partial F -test statistics for the above hypothesis testing problem as t_j and F_j , respectively, one can show:

$$t_j^2 = F_j.$$

Model with Quantitative and Qualitative Predictors

Common scenario in statistical modelling

In practice, we often encounter datasets containing:

- **Quantitative predictors:** Continuous variables (e.g., age, income, temperature)
- **Qualitative predictors:** Categorical variables (e.g., gender, treatment group, region)

General formulation

$$y = \beta_0 + \sum_{j=1}^p \beta_j x_j + \sum_{k=1}^q \alpha_k z_k + \varepsilon$$

where:

- x_j : Quantitative predictors ($j = 1, \dots, p$)
- z_k : Indicator variables for qualitative predictors ($k = 1, \dots, q$)
- $\varepsilon \sim N(0, \sigma^2)$

Important note

For a qualitative variable with m levels, we need $m - 1$ indicator variables to avoid perfect multicollinearity (dummy variable trap).

Dummy Variable Coding

A qualitative variable with 3 levels

Suppose we have a predictor "Program" with levels: Science, Arts, Business.

We create two indicator variables:

$$z_1 = \begin{cases} 1 & \text{if Arts} \\ 0 & \text{otherwise} \end{cases}$$
$$z_2 = \begin{cases} 1 & \text{if Business} \\ 0 & \text{otherwise} \end{cases}$$

Science becomes the **reference category** (when $z_1 = 0, z_2 = 0$).

$$y = \beta_0 + \beta_1 x + \alpha_1 z_1 + \alpha_2 z_2 + \varepsilon$$

Interpretation

- β_0 : Expected response for Science students when all quantitative predictors = 0
- α_1 : Difference between Arts and Science students (" + " means increase and " - " means decrease)
- α_2 : Difference between Business and Science students (" + " means increase and " - " means decrease)

Dummy Variable Coding: Numerical Example

Suppose the fitted model is

$$y = 50 + 2x + 5z_1 - 3z_2$$

where

- x = quantitative predictor (e.g., years of experience),
- $z_1 = 1$ if Arts, 0 otherwise,
- $z_2 = 1$ if Business, 0 otherwise,
- **Science** is the reference category.

Predicted values when $x = 10$

- **Science student** ($z_1 = 0, z_2 = 0$):

$$\hat{y} = 50 + 2(10) = 70$$

- **Arts student** ($z_1 = 1, z_2 = 0$):

$$\hat{y} = 50 + 2(10) + 5 = 75$$

- **Business student** ($z_1 = 0, z_2 = 1$):

$$\hat{y} = 50 + 2(10) - 3 = 67$$

Interpreting The Coefficient Estimates

- $\hat{\beta}_0 = 50$: Predicted y for a **Science student** with $x = 0$ (intercept). *“The baseline average outcome is 50 units when experience is zero for Science.”*
- $\hat{\beta}_1 = 2$: Each **one-unit increase in x** (e.g., +1 year experience) increases the **average y** by 2 units, **holding program type constant**. *“Same slope for all programs (parallel lines assumption).”*
- $\hat{\alpha}_1 = 5$: **Arts students** have predicted y that is **5 units higher** than Science students, **for the same x** . *“Positive = higher average outcome relative to reference.”*
- $\hat{\alpha}_2 = -3$: **Business students** have predicted y that is **3 units lower** than Science students, **for the same x** . *“Negative = lower average outcome relative to reference.”*

Note

The dummy coefficients $\alpha_1 = 5$ and $\alpha_2 = -3$ represent **differences relative to Science (differences in conditional means)**, not absolute levels. Science's baseline is $50 + 2x$; Arts adds +5; Business adds -3.

Example: Salary Model

Research question

How do years of experience (quantitative) and gender (qualitative) affect salary?

Model specification

Let y = Salary, x = Years of Experience, and

$$z = \begin{cases} 1 & \text{if Male} \\ 0 & \text{if Female} \end{cases}$$

Model:

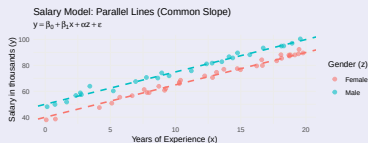
$$y = \beta_0 + \beta_1 x + \alpha z + \varepsilon$$

Interpretation

Females: $E(y|x, z = 0) = \beta_0 + \beta_1 x$

Males: $E(y|x, z = 1) = (\beta_0 + \alpha) + \beta_1 x$

Same slope β_1 for both genders (parallel lines).



When Do We Need Interaction Terms?

Key question

Does the effect of a quantitative predictor differ across levels of a qualitative predictor?

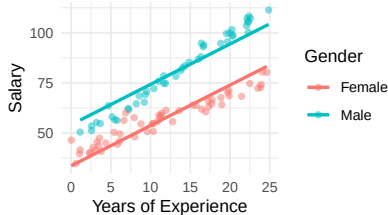
Real-world scenarios

- Does each year of experience increase salary equally for males and females?
- Does study time affect grades equally across different programs?
- Does drug dosage have the same effect for different treatment types?

Visual check

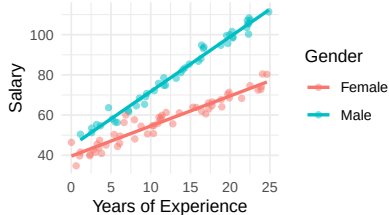
No Interaction Model

Parallel Lines: Same Slope (β_1)



Interaction Model

Non-Parallel: Different Slopes



Interaction Model Formulation

Model with interaction

Continuing our salary example:

$$y = \beta_0 + \beta_1 x + \alpha z + \gamma(x \times z) + \varepsilon$$

where $x \times z$ is the interaction term.

Interpretation by group

$$\text{Females } (z = 0) : E(y|x) = \beta_0 + \beta_1 x$$

$$\text{Males } (z = 1) : E(y|x) = (\beta_0 + \alpha) + (\beta_1 + \gamma)x$$

Coefficient meanings

- β_1 : Slope for females (reference group).
- γ : **Difference in slopes** between males and females.
- α : Intercept difference when $x = 0$.

How to Detect Need for Interaction Terms

Statistical methods

- 1 **Visual inspection:** Plot y vs x for each category.
- 2 **Hypothesis test:** Test $H_0 : \gamma = 0$ vs $H_a : \gamma \neq 0$.
- 3 **Model comparison:** Compare models with/without interaction using F-test or AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion).

F-test for interaction

$$F = \frac{(SSE_r - SSE_f)/(df_r - df_f)}{SSE_f/df_f},$$

where :

SSE_r : Sum of squares error for reduced model (no interaction)

SSE_f : Sum of squares error for full model (with interaction)

R Example: Salary Data

Simulated data and analysis

```
# Simulate data
set.seed(2509)
n <- 100
gender <- sample(c("Male", "Female"), n, replace=TRUE)
experience <- runif(n, 0, 20)
salary <- 40000 + 2000*experience + 5000*(gender=="Male") +
          300*(gender=="Male")*experience + rnorm(n, 0, 5000)

# Create indicator variable
is_male <- as.numeric(gender == "Male")

# Fit models
model1 <- lm(salary ~ experience + is_male)
model2 <- lm(salary ~ experience + is_male + experience:is_male)

# Compare models
summary(model2)
anova(model1, model2)
```

Interpreting R Output

Model summary (with interaction)

Residuals:

Min	1Q	Median	3Q	Max
-10554.5	-3503.1	-9.8	3117.2	12066.2

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
(Intercept)	39331.3	1405.5	27.984 < 2e-16 ***
experience	2062.7	118.8	17.361 < 2e-16 ***
is_male	6855.3	2058.7	3.330 0.00123 **
experience:is_male	204.2	180.2	1.133 0.26006

Interpretation

$$\widehat{\text{Salary}} = 39,331.3 + 2,062.7 \text{ Experience} + 6,855.3 \text{ Male} + 204.2 \text{ Experience} \times \text{Male}$$

- Base salary (0 experience) for females: \$39,331.3
- Female salary increase per year: \$2,062.7
- Male salary premium at 0 experience: $39,331.3 + 6,855.3 = \$46,186.6$
- Additional increase per year of experience for males compared to females is \$204.2
- Interaction p-value = 0.26006 (not significant)

Model Comparison in R

ANOVA results

```
> anova(model1, model2)
Analysis of Variance Table

Model 1: salary ~ experience + is_male
Model 2: salary ~ experience + is_male + experience:is_male
Res.Df      RSS Df Sum of Sq    F Pr(>F)
1       97 2323053190
2       96 2292403004   1  30650186 1.2836 0.2601
```

Decision making

H_0 : Model 1 without interaction (parsimonious reduced model) is preferred

H_a : Model 2 with interaction (full model) is preferred

- $F = 1.2836, p = 0.2601$
- At $\alpha = 0.05$, we don't reject $H_0 : \gamma = 0$
- Evidence for interaction is weak (not significant).
- May choose simpler model (no interaction) for parsimony.

Multiple Categories and Interactions

Qualitative variable with 3 levels

Suppose Program has 3 levels: Science (reference), Arts, Business.

Model with interactions:

$$y = \beta_0 + \beta_1 x + \alpha_1 z_1 + \alpha_2 z_2 \\ + \gamma_1(x \times z_1) + \gamma_2(x \times z_2) + \varepsilon$$

where $z_1 = \text{Art}$ and $z_2 = \text{Business}$

Interpretation

$$\text{Science: } E(y) = \beta_0 + \beta_1 x$$

$$\text{Arts: } E(y) = (\beta_0 + \alpha_1) + (\beta_1 + \gamma_1)x$$

$$\text{Business: } E(y) = (\beta_0 + \alpha_2) + (\beta_1 + \gamma_2)x$$

Practical Guidelines

Step-by-step approach

- 1 Start with main effects model (no interactions),
- 2 Check residual plots for model assumptions,
- 3 Plot y vs x for each category (look for non-parallel patterns),
- 4 Add interaction terms based on theoretical knowledge,
- 5 Test significance of interaction terms,
- 6 Use hierarchical principle: if interaction is included, keep main effects,
- 7 Compare models using F-test.

Common pitfalls

- Overfitting with too many interaction terms.
- Interpreting main effects when interaction is present.
- Forgetting the hierarchical principle.
- Not checking model assumptions after adding interactions.

Coincident Regression Lines

Coincident regression lines

Occur when different groups/categories have **identical** regression relationships with a quantitative predictor. Both the **intercepts** and **slopes** are the same across all groups.

Mathematical representation

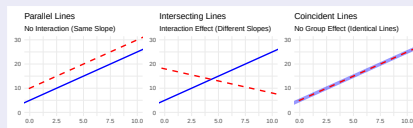
For two groups (Group A and Group B):

$$\text{Group A: } y = \beta_0 + \beta_1 x + \varepsilon$$

$$\text{Group B: } y = \beta_0 + \beta_1 x + \varepsilon$$

Identical parameters for both groups.

Visual representation



Both groups follow exactly the same line.

Key implication

If regression lines are coincident, the **qualitative predictor provides no additional information** beyond what's already explained by the quantitative predictor.

Example: Student Performance Study

Research scenario

Investigating whether **study hours** affect **exam scores** differently for **online** vs **in-person** students.

- y : Exam score (0 – 100),
- x : Study hours per week,
- z : Delivery mode (0 = In-person, 1 = Online).

Hypothesized model

Full model with interaction:

$$y = \beta_0 + \beta_1 x + \alpha z + \gamma(x \times z) + \varepsilon$$

Coincident lines occur when:

$$\alpha = 0 \quad \text{AND} \quad \gamma = 0$$

Both main effect and interaction are zero.

Interpretation of results

- If we fail to reject H_0 : Evidence for coincident lines.
- If we reject H_0 : Lines are not coincident.
- Practical implication: Delivery mode doesn't matter for the study hours-score relationship.

Example

In a study of employee productivity at a company, each employee works under one of three management styles: Style A (traditional), Style B (collaborative), or Style C (remote). The productivity score (y) of employees and their years of experience (x_1) are collected. The total number of employees in the study is $n = 32$. The full multiple linear regression model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3 + \varepsilon$$

where

$$x_2 = \begin{cases} 1, & \text{if management style B (collaborative)} \\ 0, & \text{otherwise} \end{cases}$$

$$x_3 = \begin{cases} 1, & \text{if management style C (remote)} \\ 0, & \text{otherwise} \end{cases}$$

The SSRs and SSEs for various models are:

Model	SSR	SSE
Full: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3$	15680.5	42.3
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2$	15520.8	202.0
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_5 x_1 x_3$	15670.2	52.6
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_4 x_1 x_2$	12450.3	3272.5
$y = \beta_0 + \beta_1 x_1 + \beta_3 x_3 + \beta_5 x_1 x_3$	13890.7	1832.1
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$	15430.6	292.2
$y = \beta_0 + \beta_1 x_1$	11250.4	4472.4

Example (cont.)

Question: Explain the meaning of $\beta_2, \beta_3, \beta_4$ and β_5 in the context of this business study.

β_2 and β_4 : Intercept and slope difference (Collaborative vs. Traditional)

Management style A ($x_2 = 0, x_3 = 0$), the MLR is $y = \beta_0 + \beta_1 x_1 + \varepsilon$

Management style B ($x_2 = 1, x_3 = 0$), the MLR is $y = (\beta_0 + \beta_2) + (\beta_1 + \beta_4)x_1 + \varepsilon$

- β_2 represents the difference in the experience-productivity **intercept** between collaborative management (B) and traditional management (A)
- β_4 represents the difference in the experience-productivity **slope** between the two lines of collaborative management (B) and traditional management (A)
 - ▶ If $\beta_4 > 0$: Experience has a stronger positive effect under collaborative management
 - ▶ If $\beta_4 < 0$: Experience has a weaker effect under collaborative management
 - ▶ If $\beta_4 = 0$: The effect of experience is the same for both styles

β_3 and β_5 : Intercept and slope difference (Remote vs. Traditional)

Management style A ($x_2 = 0, x_3 = 0$), the MLR is $y = \beta_0 + \beta_1 x_1 + \varepsilon$

Management style C ($x_2 = 0, x_3 = 1$), the MLR is $y = (\beta_0 + \beta_3) + (\beta_1 + \beta_5)x_1 + \varepsilon$

- β_3 represents the difference in the experience-productivity **intercept** between remote management (C) and traditional management (A)
- β_5 represents the difference in the experience-productivity **slope** between the two lines of remote management (C) and traditional management (A)

Example (cont.)

Question: Are Styles A and B Parallel?

Answer: To test if the lines for management styles A and B are parallel, we need to test if the slope for style B differs from the slope for style A. This means testing the test hypotheses

$H_0 : \beta_4 = 0$ (slopes are equal) vs. $H_a : \beta_4 \neq 0$

Full Model:

- $SSE_{full} = 42.3$
- $df_{full} = 32 - 6 = 26$

Reduced Model:

- Without β_4 i.e., dropping x_1x_2 but keeping x_1x_3 (so style C slope still differs from A)
- $SSE_{reduced} = 52.6$
- $df_{reduced} = 32 - 5 = 27$

Partial F-test:

$$\begin{aligned} F &= \frac{(SSE_{reduced} - SSE_{full}) / (df_{reduced} - df_{full})}{SSE_{full} / df_{full}} \\ &= \frac{(52.6 - 42.3) / (27 - 26)}{42.3 / 26} = \frac{10.3 / 1}{1.627} = 6.33 \end{aligned}$$

Critical value: $F_{0.05}(1, 26) = 4.23$

Decision

Since $F = 6.33 > 4.23$, we **reject** H_0 .

There is sufficient evidence to conclude that the lines for management styles A and B are not parallel.

Example (cont.)

Question: Are Styles A and B Coincident?

Answer: For lines to be coincident, they must have both the same intercept and the same slope. This means testing $H_0 : \beta_2 = \beta_4 = 0$ (both the intercept difference and slope difference between styles A and B are zero) vs H_a : at least one of $\beta_2, \beta_4 \neq 0$.

Full Model:

- $SSE_{full} = 42.3$
- $df_{full} = 26$

Reduced Model:

- Without β_2 and β_4
($y = \beta_0 + \beta_1x_1 + \beta_3x_3 + \beta_5x_1x_3$)
- $SSE_{reduced} = 1832.1$
- $df_{reduced} = 32 - 4 = 28$

Partial F-test:

$$\begin{aligned} F &= \frac{(SSE_{reduced} - SSE_{full}) / (df_{reduced} - df_{full})}{SSE_{full} / df_{full}} \\ &= \frac{(1832.1 - 42.3) / (28 - 26)}{42.3 / 26} = \frac{1789.8 / 2}{1.627} = 550.0 \end{aligned}$$

Critical value: $F_{0.05}(2, 26) = 3.37$

Decision

Since $F = 550.0 > 3.37$, we **strongly reject** H_0 .

There is significant evidence that the lines for management styles A and B are **not** coincident.

Practical Implications and Model Selection

When coincidence occurs

- **Simpler model** is preferred.
- Qualitative predictor can be **removed**.
- Final model: $y = \beta_0 + \beta_1 x + \varepsilon$.
- More parsimonious and easier interpretation.

Decision flowchart

- 1 Fit full model with interaction.
- 2 Test $H_0: \alpha = 0$ and $\gamma = 0$.
- 3 If fail to reject: Use simple linear regression.
- 4 If reject: Test individual terms.

Comparison of models

Model Type	# Parameters	Interpretation	Use When
Coincident	2	One line fits all	Groups are equivalent.
Parallel	3	Same slope, different intercepts	Additive effects.
Different slopes	4	Different relationships	Interaction present.

Important considerations

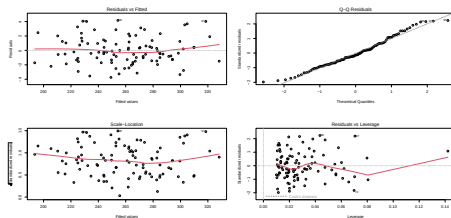
- Statistical significance vs practical significance.
- Sample size considerations.
- Domain knowledge should guide decisions.
- Always check model assumptions.

Checking Regression Assumptions

Diagnostic Plots in R:

```
par(mfrow=c(2,2))  
plot(model)
```

- 1 **Residuals vs Fitted:** Check linearity, constant variance.
- 2 **Normal Q-Q:** Check normality of residuals.
- 3 **Scale-Location:** Check homoscedasticity.
- 4 **Residuals vs Leverage:** Identify influential points.



Additional Tests:

- Shapiro-Wilk test for normality.
- Breusch-Pagan test for homoscedasticity.
- Durbin-Watson test for independence.

Coefficient of determination (r^2)

$$r^2 = \frac{SSR}{TSS} = 1 - \frac{SSE}{TSS}$$

Interpretation: Proportion of variance in y explained by model

Example from R:

$r^2 = 0.8724 = 87.24\%$ of total variation of final test grade can be explained by number of studying hours and grade of assignment.

Limitation of r^2 : Always increases when adding predictors

Adjusted coefficient of determination ($\text{adj-}r^2$)

Adjusted r^2 : penalizes adding unnecessary predictors:

$$r_{adj}^2 = 1 - \frac{SSE/(n - k - 1)}{TSS/(n - 1)} = 1 - (1 - r^2) \frac{n - 1}{n - k - 1}$$

Example from R:

$r_{adj}^2 = 0.8512$ of total variation of final test grade can be explained by the regression line.

Comparison: $r_{adj}^2 \leq r^2$, difference increases with k .

Multicollinearity

Problem: High correlation among predictors implies to make interpretation hard:

- Unstable coefficient estimates,
- Large standard errors,
- Counter intuitive signs (+ instead - and vice versa),

Variance inflation factor (VIF)

Variance Inflation Factor (VIF):

$$VIF_j = \frac{1}{1 - r_j^2}$$

where r_j^2 is the coefficient of determination from regressing X_j on other predictors.

Rule of thumb:

- $VIF < 5$ is an indicative of **no** multicollinearity.
- $5 \leq VIF < 10$ is an indicative of **minor** multicollinearity.
- $VIF \geq 10$ is an indicative of **serious** multicollinearity.

R Implementation:

```
> car::vif(model)
Hours Assignment
1.245          1.245
```

Remedies: Remove variables, combine variables, PCA, ridge regression.

Model Selection

Why model selection?

- **Too few predictors:** Underfitting, high bias.
- **Too many predictors:** Overfitting, high variance.
- **Goal:** Find the optimal balance.

The trade-off

Given k potential predictors $x_1, x_2, \dots, x_k$, we want to select the "best" subset of predictors for our regression model.

Number of possible models

With k predictors, there are:

2^k possible submodels

Including the null model (intercept only) and full model (all predictors).

Notation for a submodel

Let M be a submodel containing predictors from the set $\{x_{j_1}, x_{j_2}, \dots, x_{j_p}\}$.

$\text{SSE}(M)$ = Sum of Squares Error for model M

$$\text{MSE}(M) = \frac{\text{SSE}(M)}{n - (p + 1)}$$

Best Subset Selection Approach

Basic idea

Examine all 2^k possible models and select the best one according to some criterion.

Procedure

- 1 For each $p = 0, 1, \dots, k$:
- 2 Fit all $\binom{k}{p}$ models with p predictors,
- 3 Compute criterion for each model,
- 4 Identify the best model for each p ,
- 5 Choose among these best models.

Computational complexity

For k predictors:

- $k = 10$: $2^{10} = 1,024$ models.
- $k = 20$: $2^{20} = 1,048,576$ models.
- $k = 30$: $2^{30} > 1$ billion models.

Best subset becomes impractical for large k !

Largest r^2 Criterion

Definition of r^2

$$r^2 = \frac{\text{SSR}}{\text{TSS}} = 1 - \frac{\text{SSE}}{\text{TSS}}$$

where $\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$ is the **same for all models**.

r^2 for the null model

For the null model (intercept only):

$$\hat{y}_i = \bar{y} \quad (\text{for all } i)$$

$$\text{SSE}(\text{null}) = \sum_{i=1}^n (y_i - \bar{y})^2 = \text{TSS}$$

$$\text{SSR}(\text{null}) = 0$$

$$r_{\text{null}}^2 = 1 - \frac{\text{TSS}}{\text{TSS}} = 0$$

The null model always has $r^2 = 0$.

Adjusted r^2

Problem with ordinary r^2

$$r^2 = 1 - \frac{\text{SSE}/\text{TSS}}{(n-1)/(n-1)}$$

r^2 always increases when adding predictors, even useless ones!

Adjusted r^2 definition

$$r_{\text{adj}}^2 = 1 - \frac{\text{SSE}/(n-p-1)}{\text{TSS}/(n-1)} = 1 - \frac{\text{MSE}}{\text{TSS}/(n-1)}$$

where p = number of predictors in the model.

Properties of Adjusted r^2

- Penalizes models with more predictors.
- Can decrease when adding unhelpful predictors.
- **Selection rule:** Choose model with **largest** r_{adj}^2 .
- $r_{\text{adj}}^2 \leq r^2$, with equality only when $p = n - 1$.

Smallest MSE criterion

Definition of MSE

For a model M with p predictors:

$$\text{MSE}(M) = \frac{\text{SSE}(M)}{n - p - 1} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p - 1}$$

- Estimate of error variance σ^2 .
- Denominator: degrees of freedom for error.

Trade-off in MSE

- Adding predictors: SSE decreases but denominator decreases.
- MSE may increase if added predictors don't reduce SSE enough.
- **Selection rule:** Choose model with **smallest** MSE.

Relationship with Adjusted r^2

Since $\text{TSS}/(n - 1)$ is constant:

$$\arg \max r_{\text{adj}}^2 = \arg \min \text{MSE}$$

These two criteria select the same model!

Mallows' C_p Criterion

Motivation for C_p

We want a model that balances:

- 1 Good fit to current data (low bias).
- 2 Good performance on new data (low variance).

Definition of Mallows' C_p

For a model M with p predictors:

$$C_p = \frac{\text{SSE}_p}{\text{MSE}_f} + 2p - n,$$

where:

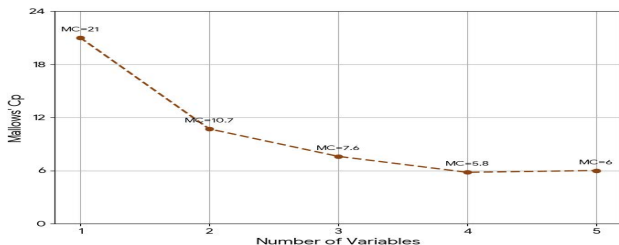
- p is the number of regression coefficients. That is $p = k$ for a model of k predictors without intercept, and $p = k + 1$ for a model with intercept.
- $\text{SSE}_p = \text{SSE}(M)$ is the sum squares errors of the submodel under consideration.
- $\text{MSE}_f = \text{MSE}(X_1, X_2, \dots, X_k)$, the mean squares errors of the full model.

Interpretation of C_p

Ideal C_p values

- If model M has no bias: $E[C_p(M)] \approx p$.
- If model M has bias: $E[C_p(M)] > p$.
- **Selection rule:** Choose model with **small C_p** that is **close to or below p** .

Plot C_p values of all submodels against the values of p for these models



Example: C_p calculation for three models

Suppose we have:

- $n = 30$ observations.
- Full model with $k = 5$ predictors, so $p = k + 1 = 6$ (assuming model with intercept).
- $\text{MSE}_f = 4.2$.

Calculate C_p for three models

Model	Predictors	SSE	p	MSE_p	C_p
M_1	x_1, x_2	150	3	$\frac{150}{27} = 5.56$	$\frac{150}{4.2} + 6 - 30 = 11.7$
M_2	x_1, x_3, x_4	130	4	$\frac{130}{26} = 5.00$	$\frac{130}{4.2} + 8 - 30 = 8.95$
M_3	x_1, x_2, x_4, x_5	115	5	$\frac{115}{25} = 4.60$	$\frac{115}{4.2} + 10 - 30 = 7.38$

Which model is best by C_p ?

- Compare C_p to p : For M_3 , $p = k + 1 = 5$, $C_p = 7.38$.
- All have $C_p > p$, suggesting some bias.
- M_3 has smallest C_p but might still be biased.

Forward Addition Procedure

Basic idea

Start with null model, add predictors one at a time based on significance.

Algorithm

- 1 Start with null model.
- 2 For each predictor x_j not in model:
- 3 Compute $F_{j|M} = \frac{\text{MSR}(x_j|M)}{\text{MSE}(x_j, M)}$.
- 4 Add predictor with largest significant F .
- 5 Repeat until no significant additions.

F-statistic formula

For adding x_j to current model M :

$$F_{j|M} = \frac{[\text{SSR}(M, x_j) - \text{SSR}(M)]/1}{\text{MSE}(M, x_j)}$$

Under H_0 : x_j has no effect given M ,
 $F_{j|M} \sim F_{1, n-p-2}$.

Example: First step

Start with null model $M_0 : y = \beta_0$:

$$F_{j|\emptyset} = \frac{\text{MSR}(x_j)}{\text{MSE}(x_j)} = \frac{\text{SSR}(x_j)/1}{\text{SSE}(x_j)/(n-2)}$$

Add x_{j1} with largest significant F .

Forward Selection: R Example

R implementation with mtcars data

```
# Forward selection example
library(MASS)
full_model <- lm(mpg ~ ., data = mtcars)
null_model <- lm(mpg ~ 1, data = mtcars)

# Forward selection
forward_model <- stepAIC(null_model,
  scope = list(lower = null_model, upper = full_model),
  direction = "forward")
```

Step-by-step output

- Step 1: Add wt ($F = 112.6$, $p < 0.0001$).
- Step 2: Add cyl ($F = 19.9$, $p = 0.0001$).
- Step 3: Add hp ($F = 14.2$, $p = 0.0009$).
- Stop: No more significant predictors at $\alpha = 0.05$.

Backward Elimination Procedure

Basic idea

Start with full model, remove predictors one at a time based on insignificance.

Algorithm

- 1 Start with full model.
- 2 For each predictor x_j in model:
- 3 Compute $t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$.
- 4 Remove predictor with smallest $|t|$ if not significant.
- 5 Repeat until all predictors significant.

Equivalent F-statistic

For removing x_j from current model M :

$$F_{j|(M \setminus x_j)} = \frac{[\text{SSR}(M) - \text{SSR}(M \setminus x_j)]/1}{\text{MSE}(M)}$$

This equals t_j^2 for testing $H_0 : \beta_j = 0$.

Backward in R

```
# Backward elimination
backward_model <- stepAIC(full_model, direction = "backward")
```

Stepwise Selection

Combining Forward and Backward

At each step, consider both:

- Adding a predictor (if significant).
- Removing a predictor (if insignificant).

Algorithm

- 1 Start with some model (often null).
- 2 Perform one forward step.
- 3 Perform one backward step.
- 4 Repeat until no changes.
- 5 Use different α levels (α to enter, α to remove).

Common practice

- Typically: α to enter $<$ α to remove
- Example: $\alpha_{\text{enter}} = 0.05$, $\alpha_{\text{remove}} = 0.10$.
- Prevents cycling between adding/removing.

Stepwise in R

```
# Stepwise selection
stepwise_model <- stepAIC(null_model,
scope = list(lower = null_model, upper = full_model),
direction = "both")
```


Comparison of Methods

1 Forward Selection:

- ▶ Start with no predictors.
- ▶ Add most significant predictor.
- ▶ Continue until no significant predictors remain.

2 Backward Elimination:

- ▶ Start with all predictors.
- ▶ Remove least significant predictor.
- ▶ Continue until all predictors significant.

3 Stepwise Selection:

- ▶ Combination of forward and backward.
- ▶ Re-evaluate included variables at each step.

Important considerations

- 1 Different methods may yield different "best" models.
- 2 Domain knowledge should guide final selection.
- 3 Always validate selected model on new data.

Method	Advantages	Disadvantages	Computational Cost
Best Subset	Examines all models	Computationally intensive	$O(2^k)$
Forward	Fast, good for large k	May miss interactions	$O(k^2)$
Backward	Starts with full picture	Can't handle $k > n$	$O(k^2)$
Stepwise	Most thorough	May overfit	$O(k^2)$

Comparing all methods on mtcars

```
# Load necessary libraries
library(olsrr)
library(leaps)

# Fit full model
full_model <- lm(mpg ~ cyl + disp + hp + drat + wt +
  qsec + vs + am + gear + carb,
  data = mtcars)

# 1. Best subset using leaps
best_subset <- regsubsets(mpg ~ ., data = mtcars, nvmax = 10)
summary(best_subset)

# 2. Plot Cp values
plot(best_subset, scale = "Cp")
abline(a = 1, b = 1, col = "red") # Cp = p line

# 3. Compare different methods
ols_step_best_subset(full_model)
ols_step_forward_p(full_model)
ols_step_backward_p(full_model)
ols_step_both_p(full_model)
```